netflix-analysis

November 3, 2025

[11]:
```python
# STEP 1 : Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[12]:
```python
# STEP 2 : Upload and Load Dataset

df = pd.read_csv('netflix_titles.csv')
df.head()
```

[12]:
```
  show_id     type                 title         director  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show         Blood & Water              NaN
2      s3  TV Show             Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show          Kota Factory              NaN

                                              cast        country  \
0                                              NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…            NaN
3                                              NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

           date_added  release_year rating   duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                         listed_in  \
0                                    Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                           Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …
```

```
                                            description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[13]:
```python
# STEP 3  : Basic Info
print('Rows and Columns:', df.shape)
print('\nMissing Values:\n', df.isnull().sum())
print('\nData Types:\n', df.dtypes)
```

```
Rows and Columns: (8807, 12)

Missing Values:
 show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64

Data Types:
 show_id          object
type            object
title           object
director        object
cast            object
country         object
date_added      object
release_year     int64
rating          object
duration        object
listed_in       object
description     object
dtype: object
```

[14]:
```python
# STEP 4  : Data Cleaning
df.drop_duplicates(inplace=True)
```

```python
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna('Unknown', inplace=True)

df.head()
```

```
/tmp/ipython-input-257665853.py:4: FutureWarning: A value is trying to be set on
a copy of a DataFrame or Series through chained assignment using an inplace
method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  df['country'].fillna('Unknown', inplace=True)
/tmp/ipython-input-257665853.py:5: FutureWarning: A value is trying to be set on
a copy of a DataFrame or Series through chained assignment using an inplace
method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  df['rating'].fillna('Unknown', inplace=True)
```

[14]:

```
  show_id     type                  title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                cast        country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…        Unknown
3                                                NaN        Unknown
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India
```

```
  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                           listed_in  \
0                                      Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                         description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[38]:
```python
df.drop_duplicates(inplace=True)
```

[39]:
```python
df.head()
```

[39]:
```
  show_id     type                 title         director  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show         Blood & Water              NaN
2      s3  TV Show             Ganglands  Julien Leclercq
3      s4  TV Show Jailbirds New Orleans              NaN
4      s5  TV Show          Kota Factory              NaN

                                                cast        country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…        Unknown
3                                                NaN        Unknown
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                           listed_in  \
```

```
0                                          Documentaries
1     International TV Shows, TV Dramas, TV Mysteries
2   Crime TV Shows, International TV Shows, TV Act…
3                               Docuseries, Reality TV
4   International TV Shows, Romantic TV Shows, TV …

                                         description  year_added  \
0  As her father nears the end of his life, filmm…      2021.0
1  After crossing paths at a party, a Cape Town t…      2021.0
2  To protect his family from a powerful drug lor…      2021.0
3  Feuds, flirtations and toilet talk go down amo…      2021.0
4  In a city of coaching centers known to train I…      2021.0

   duration_int    Year
0          90.0  2021.0
1           2.0  2021.0
2           1.0  2021.0
3           1.0  2021.0
4           2.0  2021.0
```

[40]:
```python
df['country'].value_counts().head(10)
```

[40]:
```
country
United States     2818
India              972
Unknown            831
United Kingdom     419
Japan              245
South Korea        199
Canada             181
Spain              145
France             124
Mexico             110
Name: count, dtype: int64
```

[41]:
```python
df['type'].value_counts()
```

[41]:
```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

[ ]:
```python

```

[15]:
```python
df['listed_in'].fillna('Unknown', inplace=True)
df['year_added'] = df['date_added'].dt.year
```

```python
df['duration_int'] = df['duration'].str.replace('min','').str.
↪replace('Season','').str.replace('s','')
df['duration_int'] = pd.to_numeric(df['duration_int'], errors='coerce')
df.head()
```

```
/tmp/ipython-input-2560945289.py:1: FutureWarning: A value is trying to be set
on a copy of a DataFrame or Series through chained assignment using an inplace
method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  df['listed_in'].fillna('Unknown', inplace=True)
```

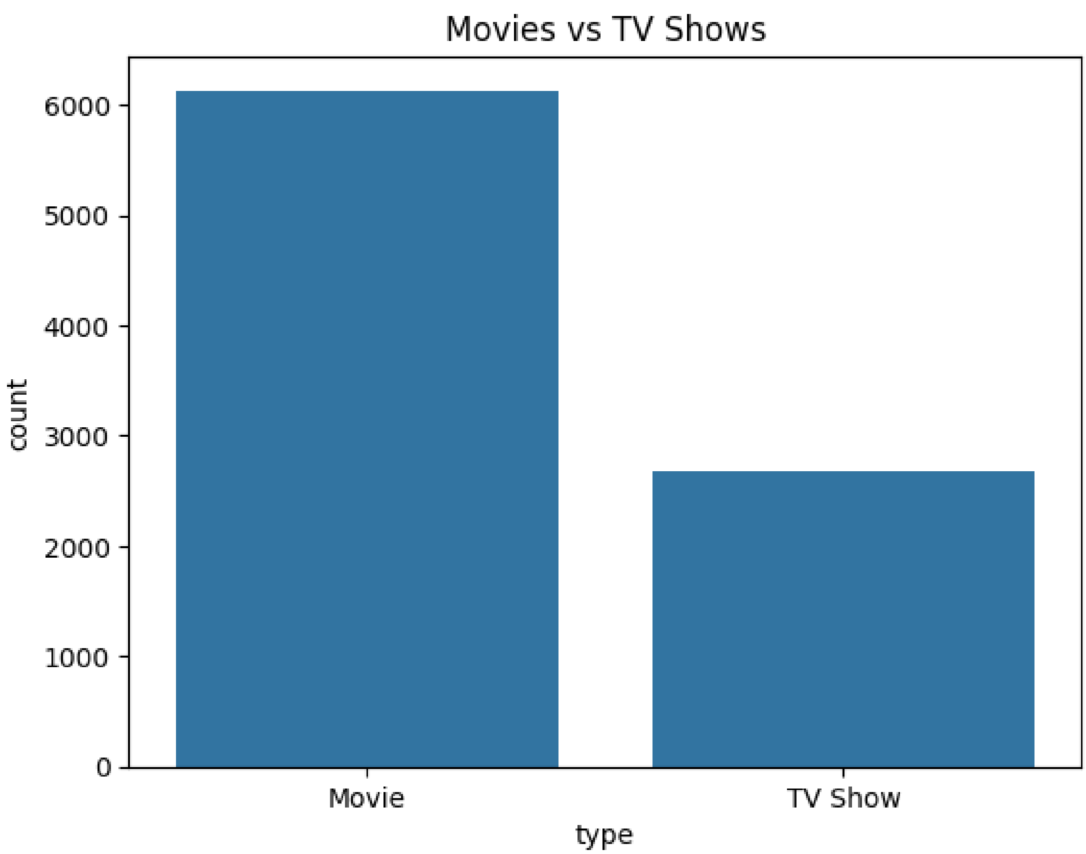
[15]:
```
  show_id     type                 title         director  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show         Blood & Water              NaN
2      s3  TV Show             Ganglands  Julien Leclercq
3      s4  TV Show Jailbirds New Orleans              NaN
4      s5  TV Show          Kota Factory              NaN

                                                cast        country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…        Unknown
3                                                NaN        Unknown
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1   International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                            Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …
```

```
                                          description  year_added  duration_int
0  As her father nears the end of his life, filmm…      2021.0          90.0
1  After crossing paths at a party, a Cape Town t…      2021.0           2.0
2  To protect his family from a powerful drug lor…      2021.0           1.0
3  Feuds, flirtations and toilet talk go down amo…      2021.0           1.0
4  In a city of coaching centers known to train I…      2021.0           2.0
```

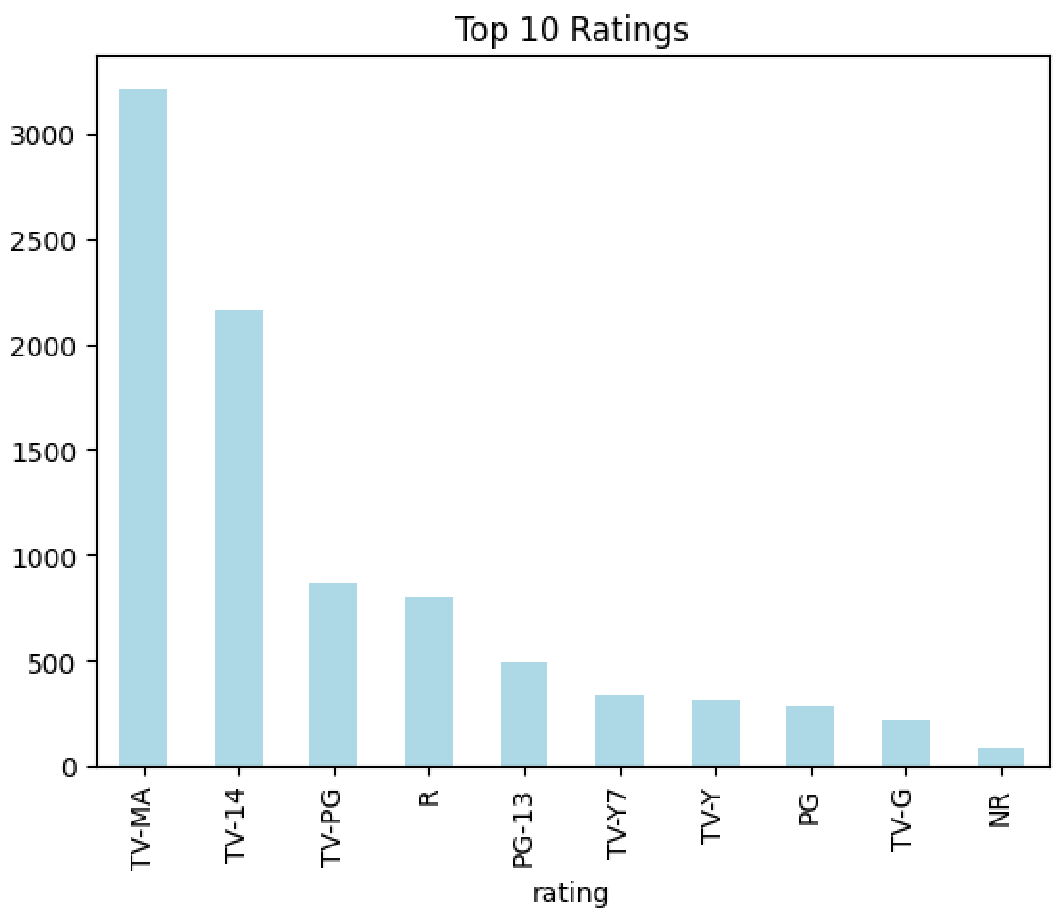
[16]:
```python
# STEP 5 : Quick Summary
print('Total Movies:', len(df[df['type']=="Movie"]))
print('Total TV Shows:', len(df[df['type']=="TV Show"]))
print('\nAverage Movie Duration:', round(df['duration_int'].mean(),2))
```

```
Total Movies: 6131
Total TV Shows: 2676

Average Movie Duration: 69.85
```

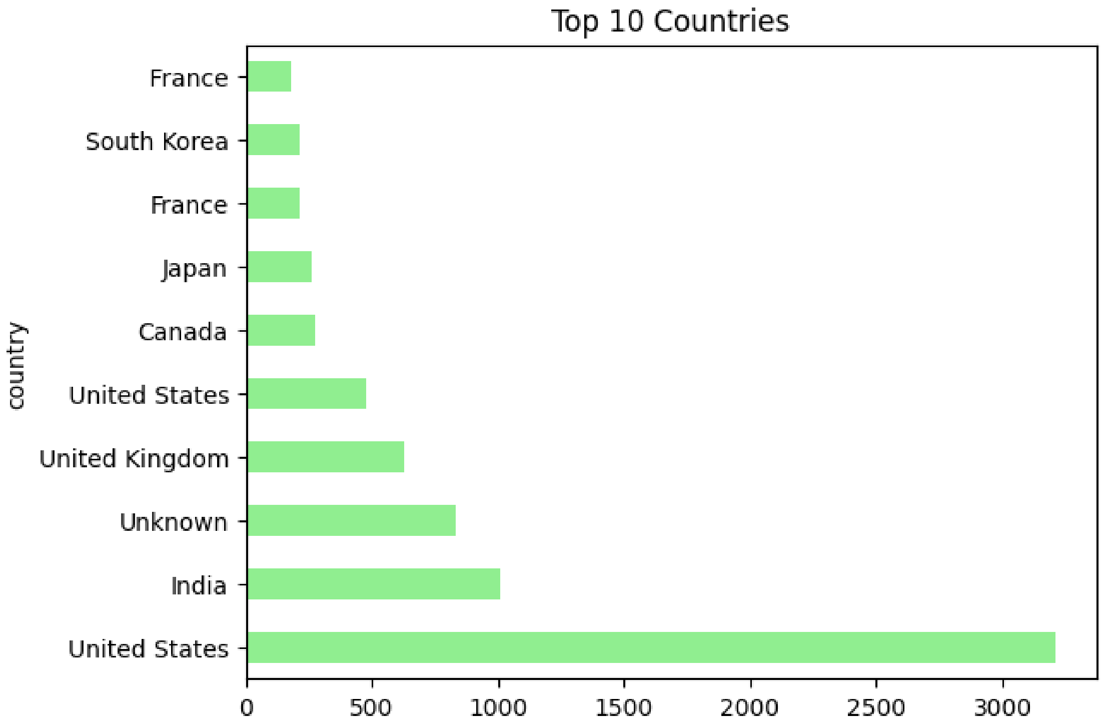
[17]:
```python
# STEP 6 : Visualizations

# 1 Movies vs TV Shows
sns.countplot(x='type', data=df)
plt.title('Movies vs TV Shows')
plt.show()
```



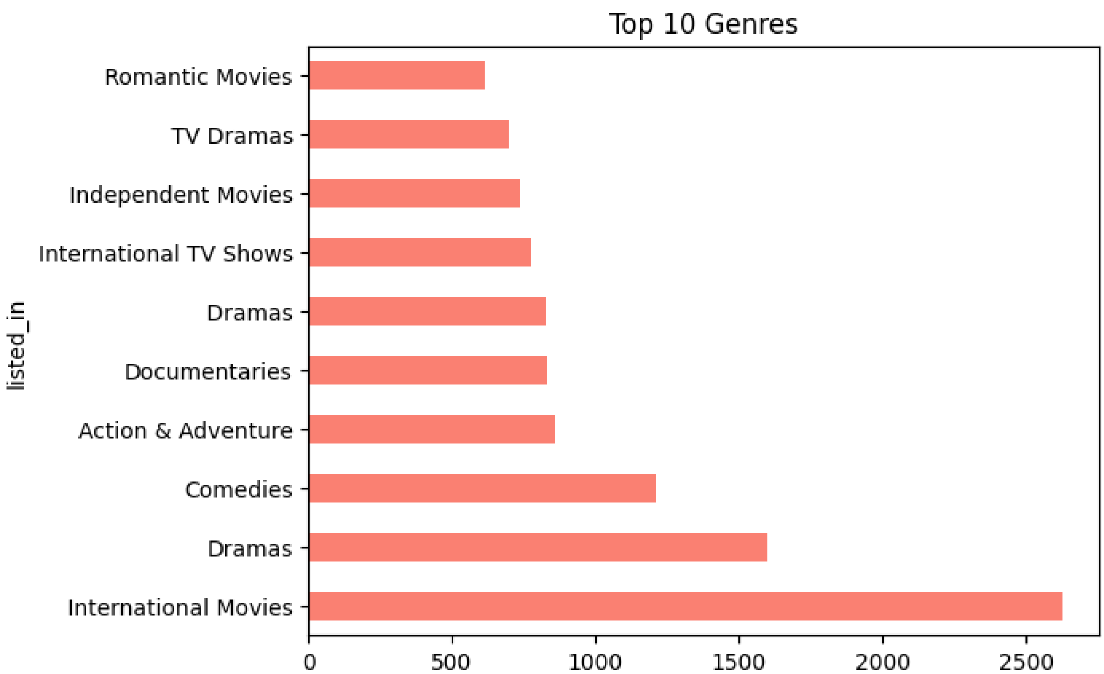
[18]:
```python
# 2 Ratings
df['rating'].value_counts().head(10).plot(kind='bar', color='lightblue')
plt.title('Top 10 Ratings')
plt.show()
```



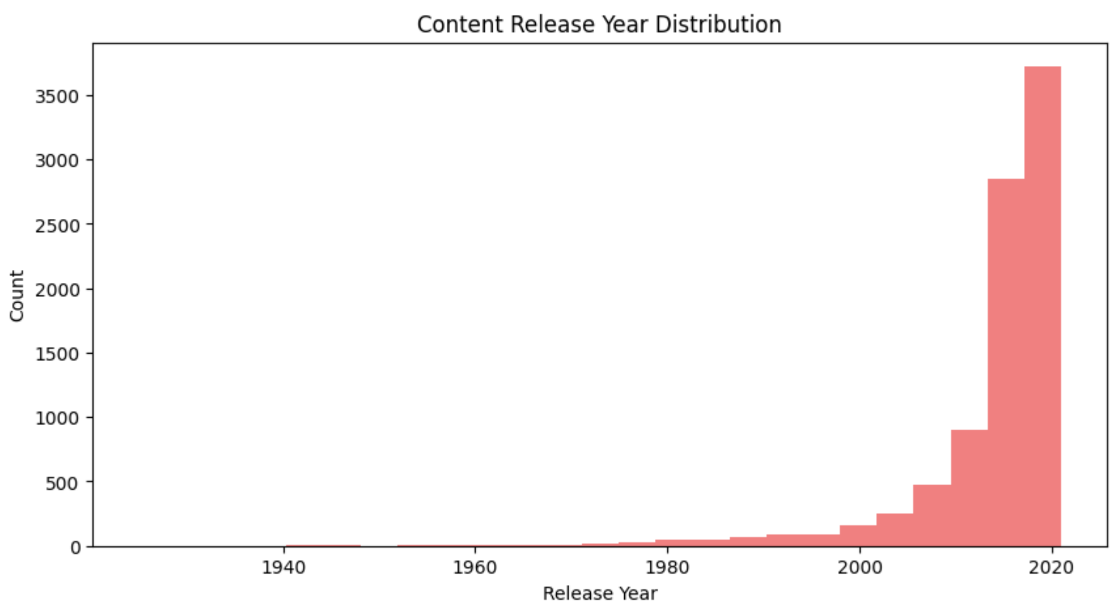
[19]:
```
# 3 Top 10 Countries
df['country'].str.split(',').explode().value_counts().head(10).
 ↪plot(kind='barh', color='lightgreen')
plt.title('Top 10 Countries')
plt.show()
```



[20]:
```
# 4 Top 10 Genres
df['listed_in'].str.split(',').explode().value_counts().head(10).
 ↪plot(kind='barh', color='salmon')
plt.title('Top 10 Genres')
plt.show()
```

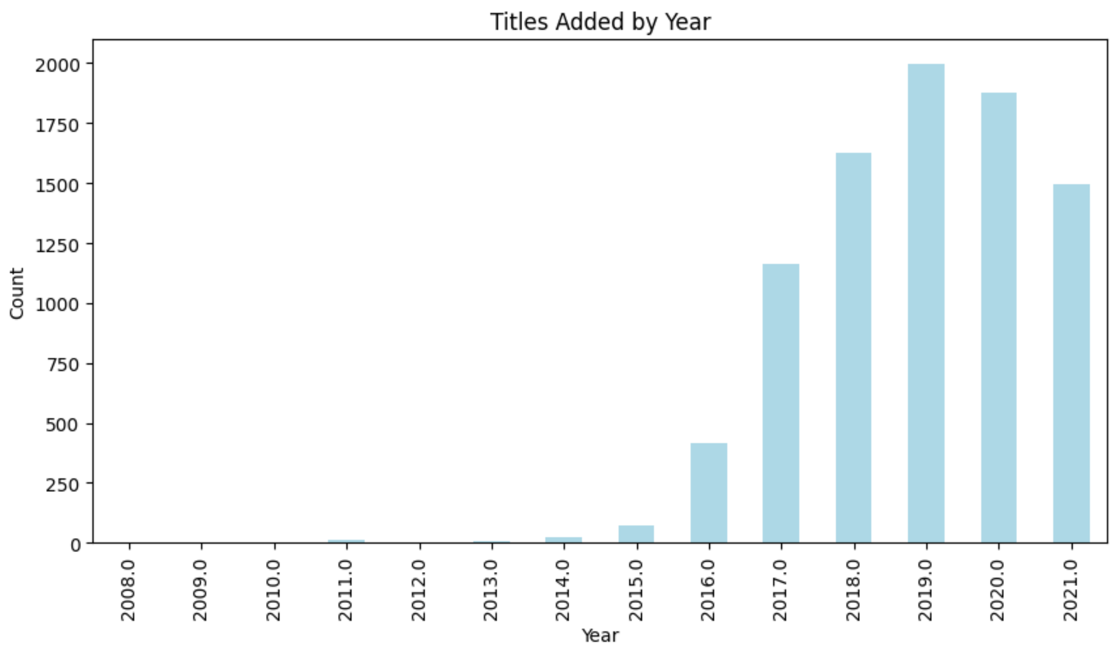


[21]:
```python
# 5 Release Year Distribution
plt.figure(figsize=(10,5))
plt.hist(df['release_year'], bins=25, color='lightcoral')
plt.title('Content Release Year Distribution')
plt.xlabel('Release Year')
plt.ylabel('Count')
plt.show()
```



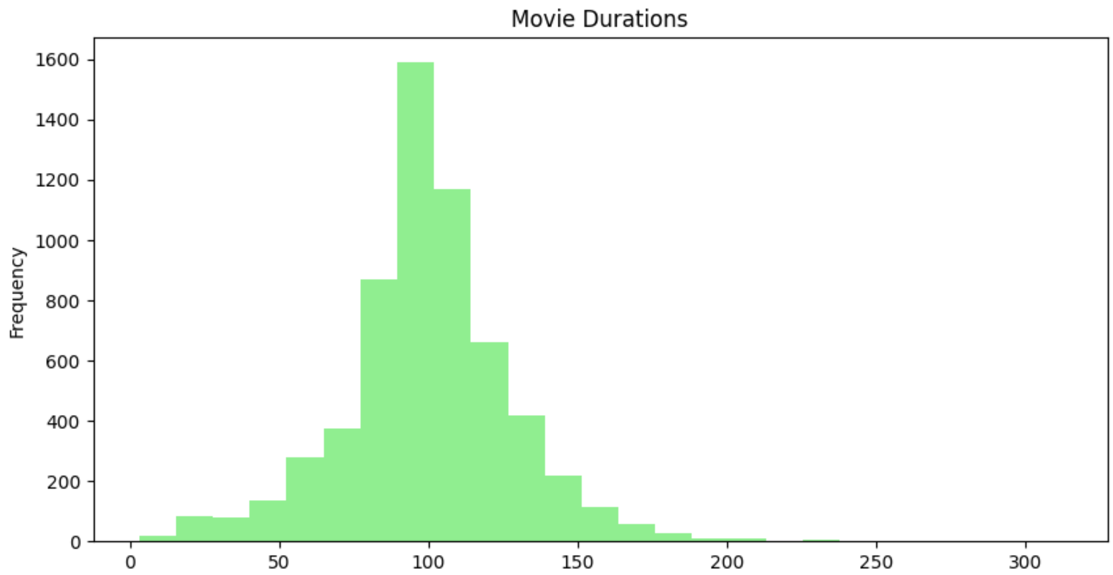
[22]:
```
# 6 Titles Added by Year
plt.figure(figsize=(10,5))
df['year_added'].value_counts().sort_index().plot(kind='bar', color='lightblue')
plt.title('Titles Added by Year')
plt.xlabel('Year')
plt.ylabel('Count')
```

[22]: Text(0, 0.5, 'Count')



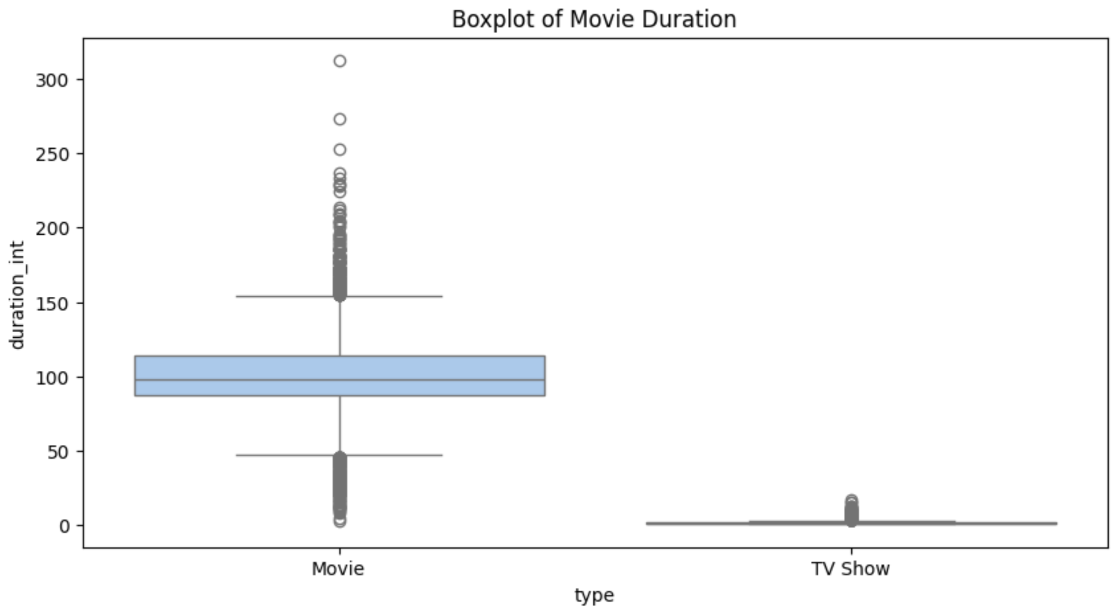
[23]:
```
# 7 Movie Durations
plt.figure(figsize=(10,5))
df[df['type'] == 'Movie']['duration_int'].plot(kind='hist', bins=25,␣
↪color='lightgreen')
plt.title('Movie Durations')
```

[23]: Text(0.5, 1.0, 'Movie Durations')

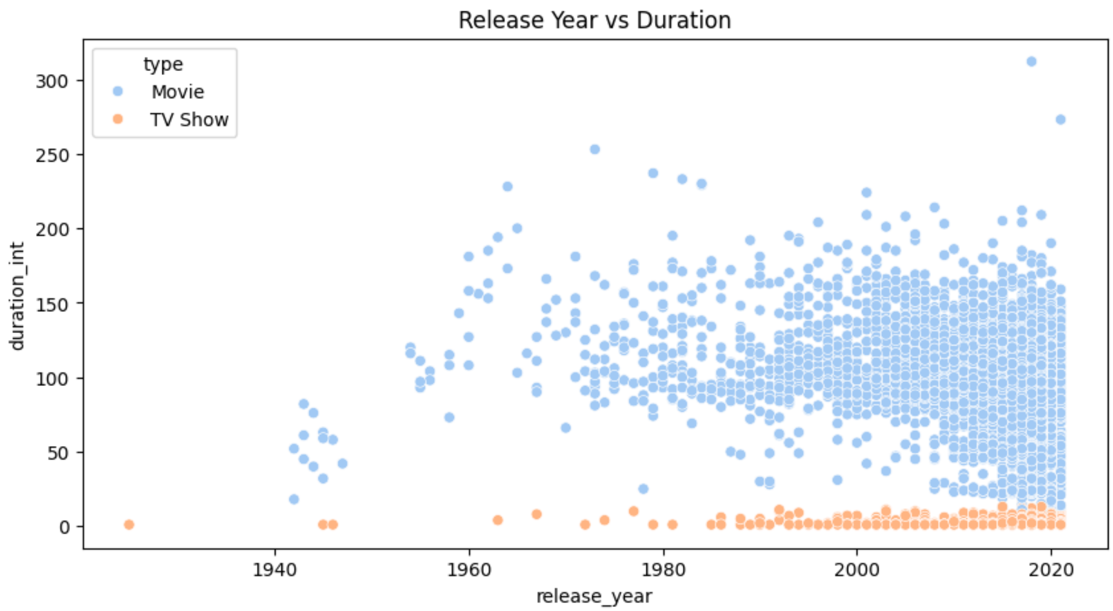


[24]:
```
# 8 Boxplot of Movie Duration
plt.figure(figsize=(10,5))
sns.boxplot(x='type', y='duration_int', data=df, palette='pastel')
plt.title('Boxplot of Movie Duration')
plt.show()
```

```
/tmp/ipython-input-2930328451.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(x='type', y='duration_int', data=df, palette='pastel')
```

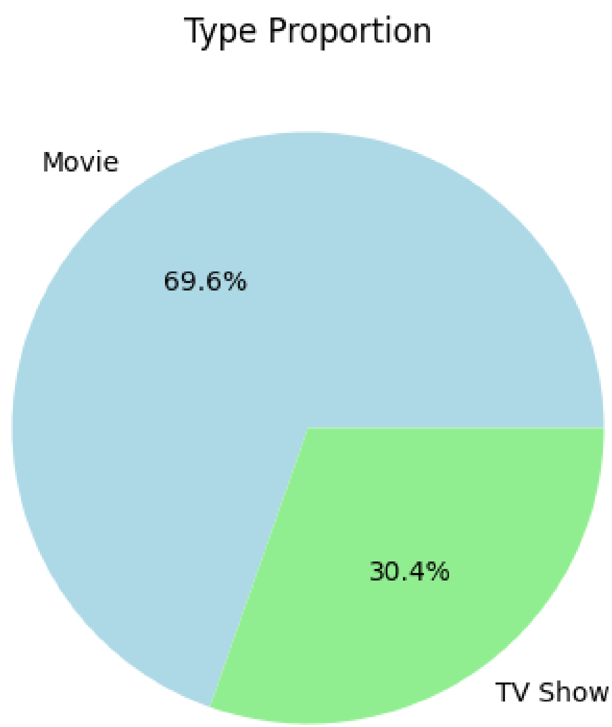


[25]:
```
# 9 Scatter Plot: Release Year vs Duration
plt.figure(figsize=(10,5))
sns.scatterplot(x='release_year', y='duration_int', data=df, hue='type',
    palette='pastel')
plt.title('Release Year vs Duration')
plt.show()
```

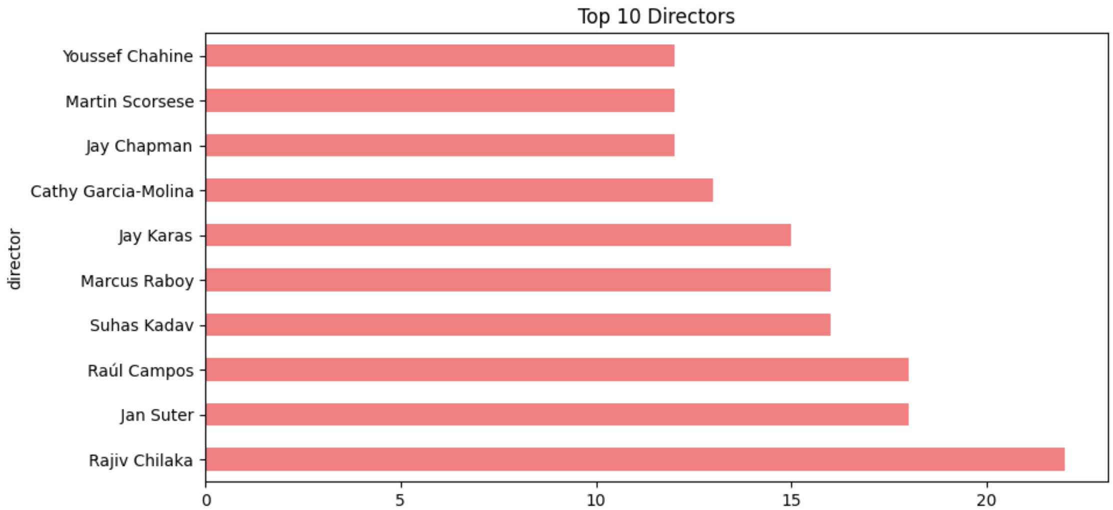


[26]:
```
# 10 Pie Chart: Type Proportion
plt.figure(figsize=(10,5))
df['type'].value_counts().plot(kind='pie', autopct='%1.1f%%',␣
↪colors=['lightblue', 'lightgreen'])
plt.title('Type Proportion')
plt.ylabel('')
```

[26]:
```
Text(0, 0.5, '')
```

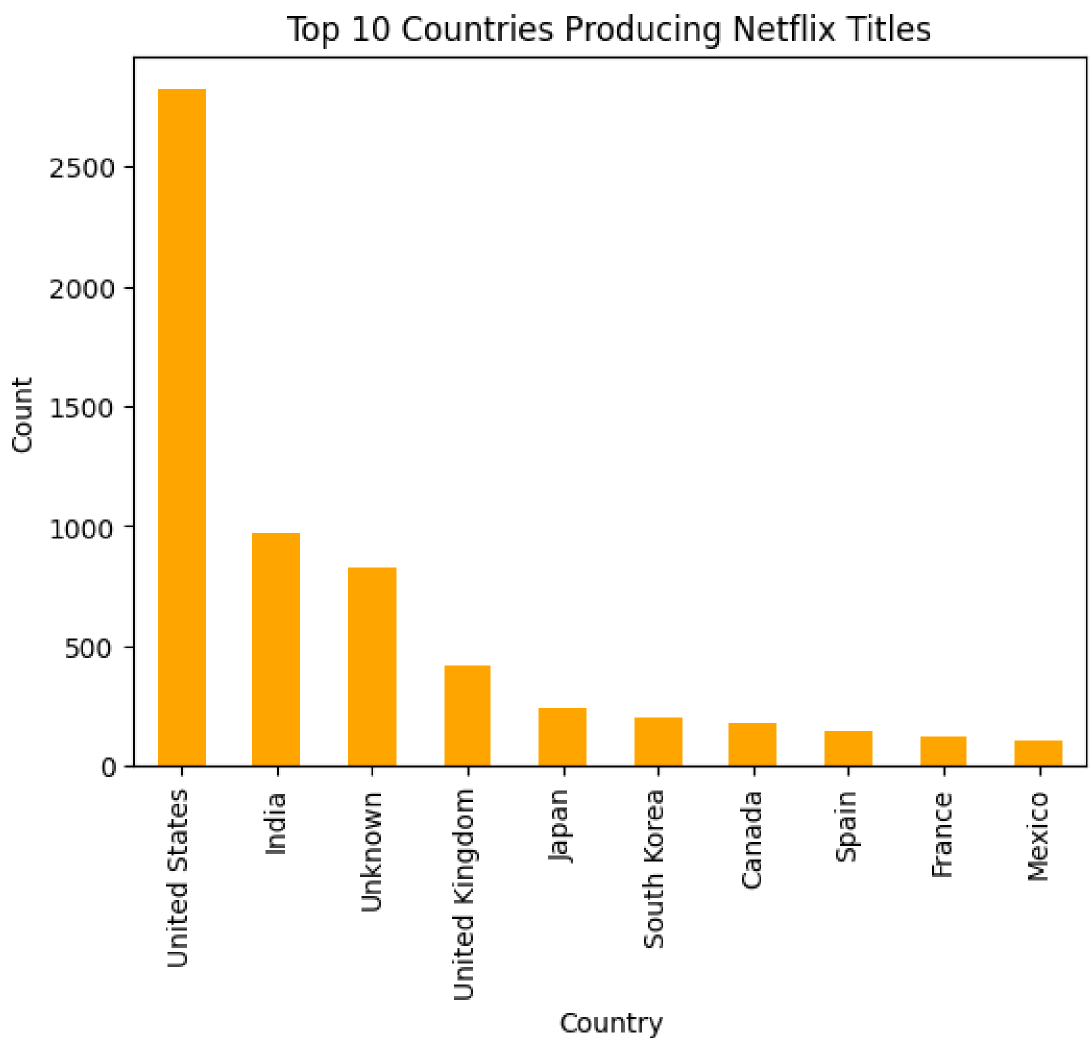


[27]:
```
# 11 Top 10 Directors
plt.figure(figsize=(10,5))
df['director'].str.split(',').explode().value_counts().head(10).
↪plot(kind='barh', color='lightcoral')
plt.title('Top 10 Directors')
plt.show()
```



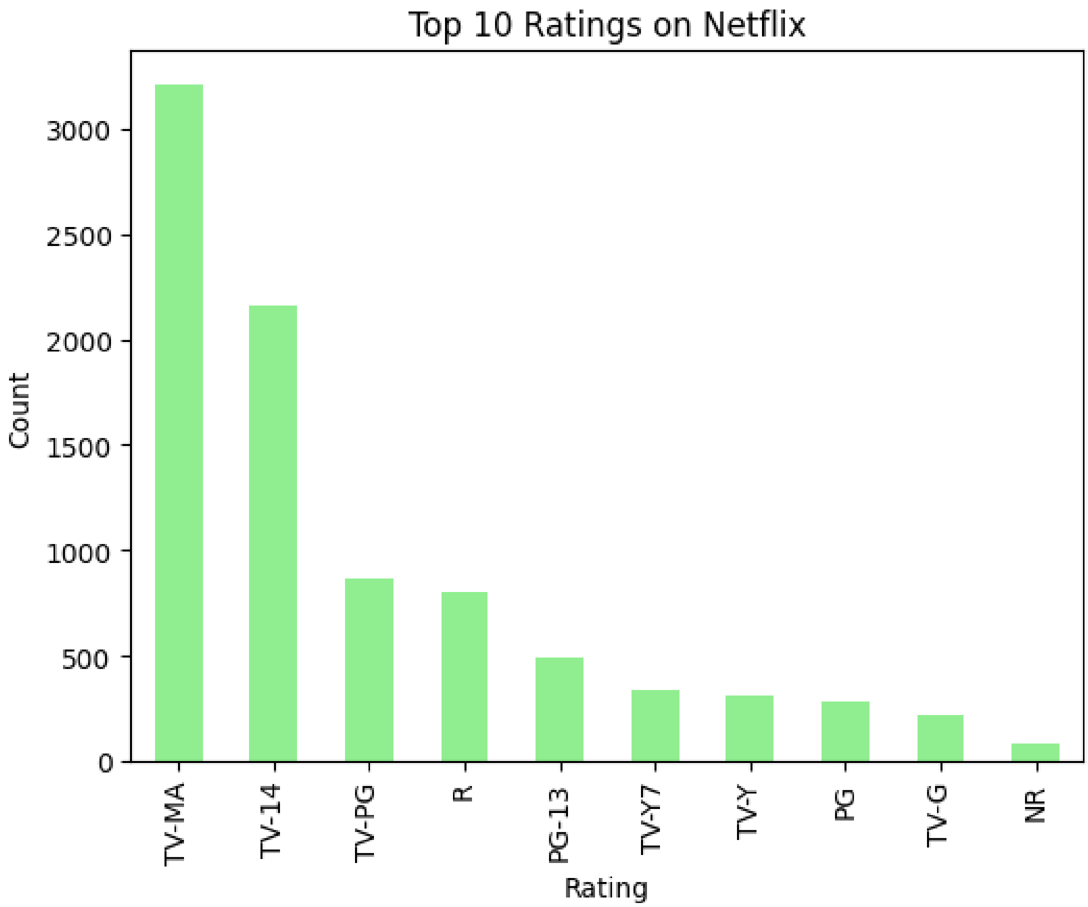
[30]:

```
#12. Top 10 Countries by Number of Titles
df['country'].value_counts().head(10).plot(kind='bar', color='orange')
plt.title("Top 10 Countries Producing Netflix Titles")
plt.xlabel("Country")
plt.ylabel("Count")
plt.show()
```

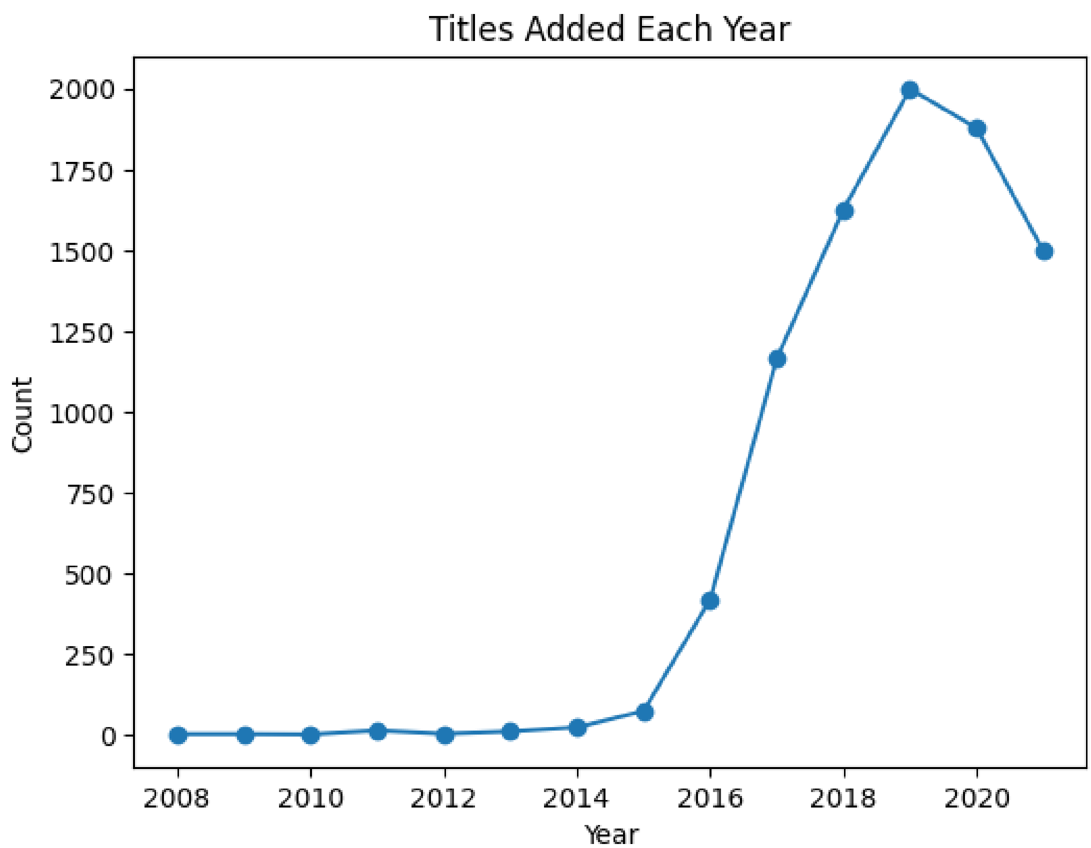


[31]:

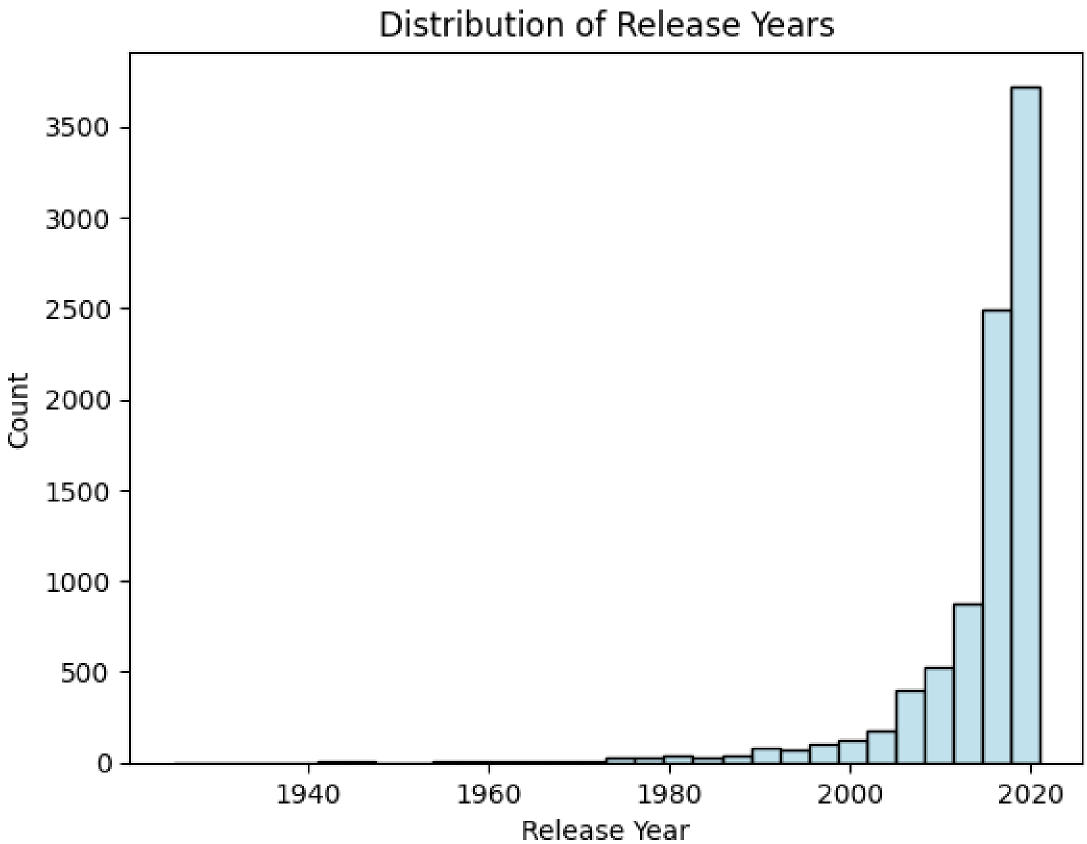
```
# 13. Ratings Distribution
df['rating'].value_counts().head(10).plot(kind='bar', color='lightgreen')
plt.title("Top 10 Ratings on Netflix")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



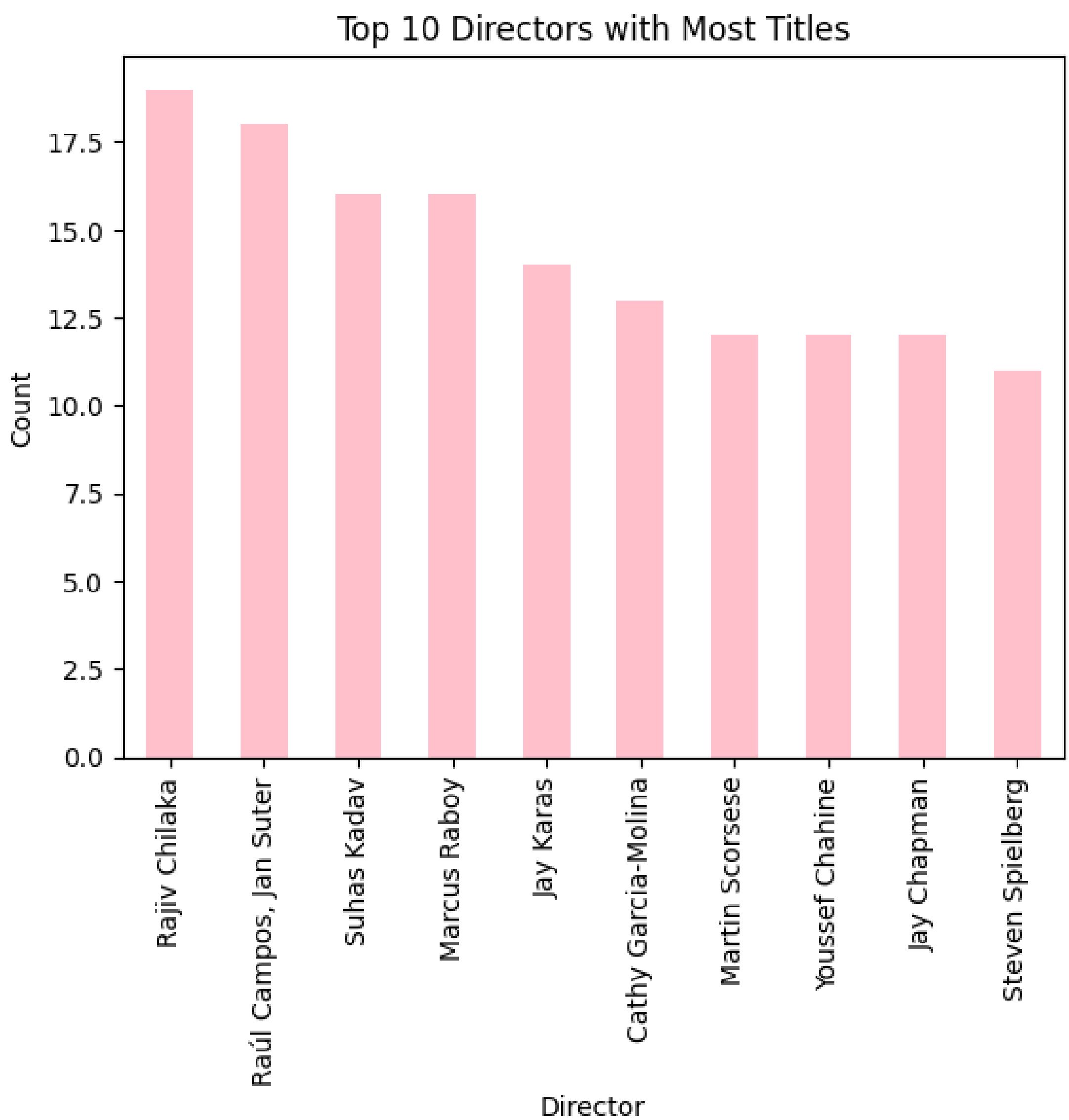
[32]:
```
#14. Content Added per Year
df['date_added'] = pd.to_datetime(df['date_added'])
df['Year'] = df['date_added'].dt.year
df['Year'].value_counts().sort_index().plot(kind='line', marker='o')
plt.title("Titles Added Each Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



[33]:
```
# 15. Movie Release Year Distribution
sns.histplot(df['release_year'], bins=30, color='lightblue')
plt.title("Distribution of Release Years")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()
```

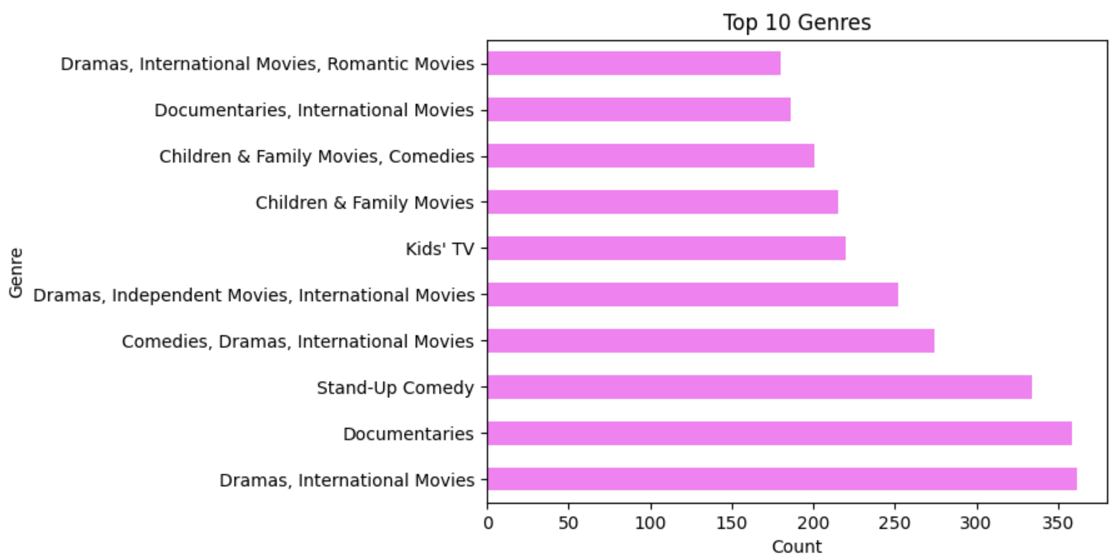


[34]:
```
#16. Top 10 Directors
df['director'].value_counts().head(10).plot(kind='bar', color='pink')
plt.title("Top 10 Directors with Most Titles")
plt.xlabel("Director")
plt.ylabel("Count")
plt.show()
```

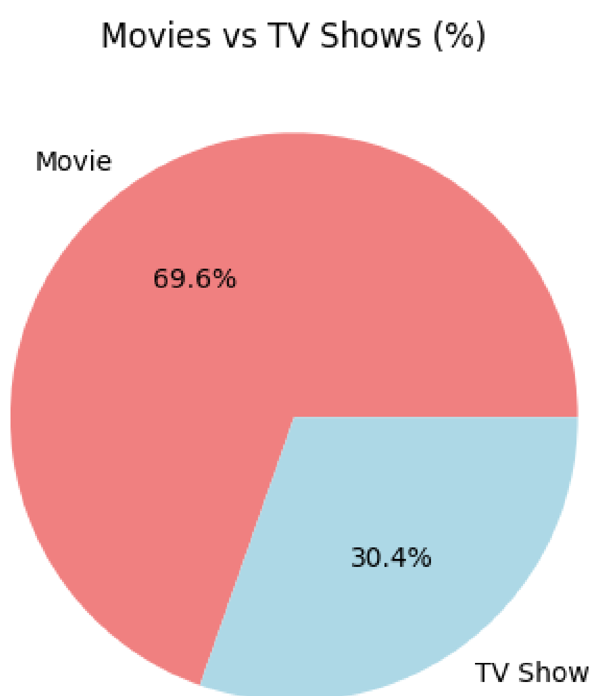


[35]:

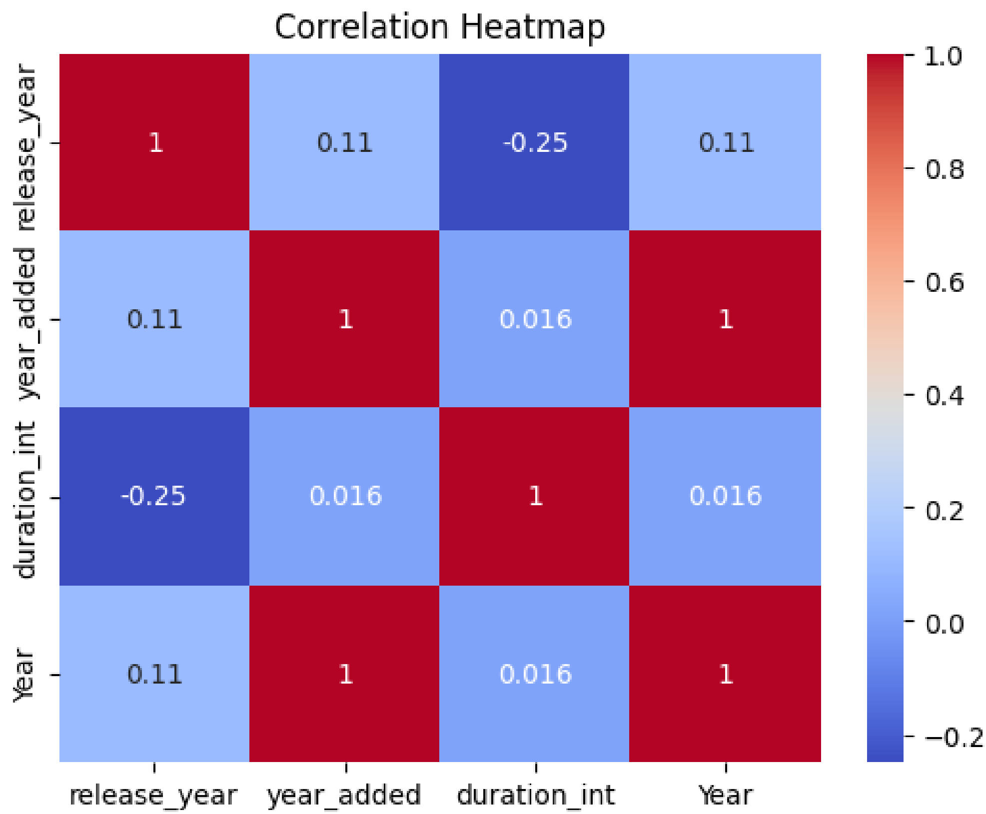
```
#17. Most Common Genres
df['listed_in'].value_counts().head(10).plot(kind='barh', color='violet')
plt.title("Top 10 Genres")
plt.xlabel("Count")
plt.ylabel("Genre")
plt.show()
```



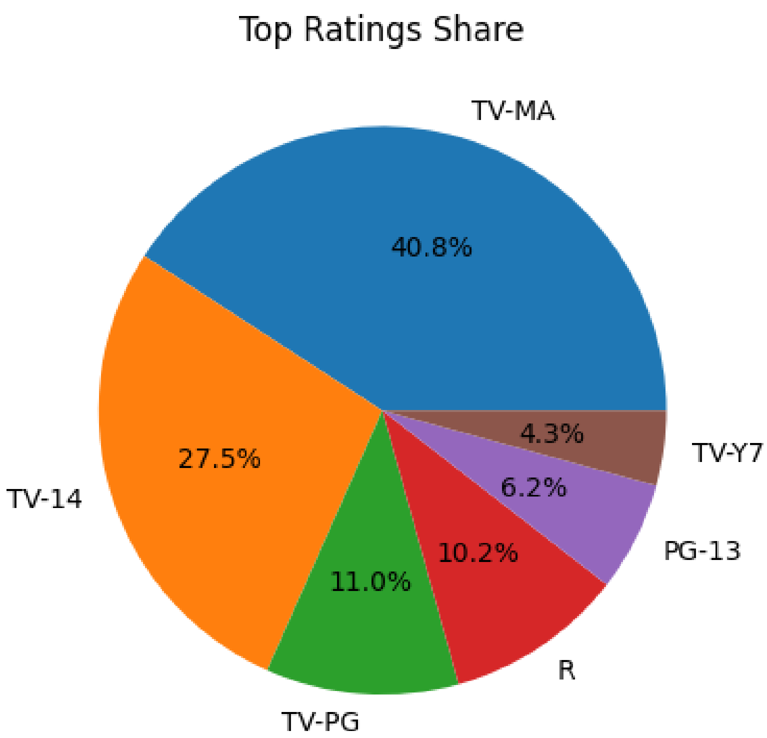
[37]:
```
#18. Pie Chart - Movies vs TV Shows
df['type'].value_counts().plot.pie(autopct='%1.1f%%',␣
↪colors=['lightcoral','lightblue'])
plt.title("Movies vs TV Shows (%)")
plt.ylabel('')
plt.show()
```



[43]:
```
#19. Heatmap

sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



[46]:
```
#20. Ratings Pie Chart
df['rating'].value_counts().head(6).plot.pie(autopct='%1.1f%%')
plt.title("Top Ratings Share")
plt.ylabel('')
plt.show()
```



[47]:
```python
# STEP 7 : Insights Summary
print(' Simple Insights:')
print('1. Movies are more than TV Shows.')
print('2. Most content is from the United States, India, and UK.')
print('3. TV-MA and TV-14 are the most common ratings.')
print('4. Most movies are between 90-120 minutes long.')
print('5. Big rise in content after 2015.')
print('6. Popular genres: Dramas, International Movies, Comedies.')
```

```
 Simple Insights:
1. Movies are more than TV Shows.
2. Most content is from the United States, India, and UK.
3. TV-MA and TV-14 are the most common ratings.
4. Most movies are between 90-120 minutes long.
5. Big rise in content after 2015.
6. Popular genres: Dramas, International Movies, Comedies.
```

[ ]: